

The Reciprocal Time Window as a Precondition of the Recognition Obligation

Article 41 of Additional Protocol I, customary international humanitarian law, and the temporal preconditions of recognising an expressed intention

5 August 2026 · Version 6, 27 August 2026 · CC BY 4.0 · DOI: 10.5281/zenodo.21224900

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Thesis. Recognition of an intention requires time on both sides of the exchange. This note calls that interval the reciprocal time window. ‘Reciprocal’ does not mean that the legal obligation itself is reciprocal: the duty analysed here binds a party regardless of the conduct of its adversary. The term refers only to the two-sided temporal structure of the event. One party must be able to express the intention; the other must be able to perceive it and act on it. Because the obligation to recognise a person *hors de combat* already exists under customary international humanitarian law and, for States parties, Article 41 of Additional Protocol I, the minimum window required for recognition is a structural precondition for complying with that obligation in practice. Whether a system preserves that precondition can be tested and must be examined in any legal review applicable to the weapon, including under Article 36 of Additional Protocol I where that provision applies and under corresponding national review requirements where it does not. Compressing the window below the minimum required for the exchange is therefore an examinable design decision, not a technical circumstance.

1. Purpose and placement

This note completes a documented progression across two earlier instruments. First, the publication of HF SIGNAL 01 — a surrender-recognition specification with a passive visual modality and an active electronic modality using two conjoined channels (a 7 Hz pulsating white light and a published 144.0 MHz FM reference value, subject to regulatory re-examination) — removed the categorical claim that no machine-readable surrender signal can be publicly specified: such a candidate now exists in public, technical form. It does not, and does not purport to, establish operational feasibility in every environment; occlusion, interference, spoofing, sensor performance, and classification remain open engineering questions. Second, the accompanying working paper (‘Paper III’) advanced the declared-choice thesis: once the specification is public, non-recognition ceases to be a limitation and becomes an examinable decision. This note takes the third step. Where the legal obligation to recognise already exists, the examination has a defined object: the time window that makes recognition possible. That window is a structural precondition for the obligation to operate in practice, and whether it is preserved is therefore a matter for legal review.

This note does not repeat Paper III. Paper III addressed the status of non-recognition: limitation or decision. This note addresses the status of the time window itself: technical circumstance or reviewable design parameter. The two arguments are cumulative, not duplicative.

2. Recognition as a reciprocal act

The element introduced here, and not developed in the earlier instruments, is the reciprocal structure of recognition. Recognising an intention is not a unilateral reading of a signal. At a minimum, the exchange contains three temporally ordered steps: (i) emission of the signal by the party expressing the protected intention; (ii) perception and evaluation of the signal by the opposing system or operator; and (iii) translation of that evaluation into attack inhibition — withholding or aborting the engagement — before the point of irreversibility. Each step takes time. The exchange therefore has an irreducible minimum duration: an end-to-end interval running from emission to inhibition. Here, ‘reciprocity’ does not mean

reciprocity of legal obligation. Surrender is a unilateral manifestation to which a corresponding duty of the opposing party attaches. The term describes only the two-sided temporal structure of expression and recognition: the interval must accommodate both the person's act of making the intention perceptible and the opposing system's act of perceiving it and responding to it. Throughout this note, the point of irreversibility is a property of the causal sequence itself — the point after which no intervention can alter the outcome — rather than a function of any particular capacity to respond (see 'Systemic Arrestability', Section 3.2).

A fourth element — a response perceivable by the person expressing the intention, confirming that the signal has been received — completes the exchange in operational terms and is decisive for the reliability of the interaction. This note does not treat that response as a precondition of the legal obligation, which is discharged by withholding attack itself. It remains a design and reliability requirement. The legal argument rests only on the end-to-end segment from expression to attack inhibition.

The central distinction follows. Below the minimum duration of the exchange, recognition is not merely missed; it is precluded. A missed recognition is a failure within a process that could have succeeded. A precluded recognition is different: the process itself has been eliminated. The legal consequences therefore differ. One scope limitation follows directly. The protected condition is not limited to surrender: a person is also protected when in the power of an adverse party, or when incapacitated by wounds or sickness and therefore incapable of defending himself. The reciprocal window analysed here concerns only the limb that turns on an expressed intention, because only that limb involves an act of emission. The other limbs raise a different problem — recognition of a state rather than recognition of an expression — which this note does not address and to which the analysis should not be extended without further argument.

The relevant threshold is not a fixed number of milliseconds. It is the scenario-specific end-to-end interval needed for a protected intention to be expressed, detected, evaluated, and translated into attack inhibition, with the required degree of reliability, before the engagement becomes irreversible. Its adequacy depends at least on the duration and observability of the signal; sensor coverage and performance; detection and classification latency; the confidence threshold applied; the time required to inhibit or abort the effector; and the effector's point of irreversibility. The threshold is therefore a measurable and verifiable parameter, not a tautology.

Measurability, however, comes with a qualification. If a system's behaviour changes over its operational life — for example because it adapts, learns, or is updated after review — the interval is not a single fixed value but a distribution that may shift after measurement. Two consequences follow. First, the parameter to be recorded is the reasonably foreseeable adverse case within the system's validated operational envelope, not the typical case. Second, the determination cannot be treated as a one-off act: once the system's relevant behaviour changes, the earlier verification no longer establishes the same property. The requirement, in the machine-safety context, that systems exhibiting self-evolving behaviour remain correctable at all times in order to maintain inherent safety points in the same direction.

3. The window as a precondition of exercisability

Article 41 of Additional Protocol I prohibits attacking a person who is recognised or who, in the circumstances, should be recognised as *hors de combat*. The ICRC Commentary on the Additional Protocols (1987) interprets the second limb by reference to the standard of a reasonable combatant acting in good faith in the circumstances ruling at the time: what 'should have been recognised' is assessed against what such a person could have perceived. That standard assumes that perception remains possible. If an engagement cycle is compressed below the minimum duration required for the exchange, the problem is not simply that the evidentiary threshold becomes harder to meet. The factual basis on which the standard operates has been removed.

This note therefore does not claim that the time window forms part of the treaty content of Article 41. It makes a narrower, and structurally stronger, claim: the window is the precondition without which the obligation cannot be exercised in practice, and preconditions of exercisability are precisely what a legal review of the weapon can examine. A party that designs, procures, or deploys a system whose engagement cycle forecloses the exchange does not escape the norm; it renders compliance with the recognition obligation unverifiable by design.

Two objections define the limits of the claim. The first is that Article 41 prohibits attacking a person who has surrendered but does not require an attacker to slow an attack so that an adversary may surrender. On that reading, no duty to provide a window can be derived from the provision. The objection is well taken, but it addresses a claim this note does not make. Nothing here asserts a right to a pause. The relevant distinction is between a surrender manifested too late in a particular engagement — a failure within a process that could have succeeded — and a system configured so that no otherwise timely manifestation, made within the intended circumstances of use, can be recognised before irreversibility. The first is a contingency of combat. The second is a property of the design. Only the second is placed before legal review.

The second objection is the mirror image of the first, and it is more serious: where acceptance of surrender is not feasible in the circumstances, the surrender does not bind the attacking party, so a compressed engagement cycle defeats the surrender rather than exposing a defect in the system. That proposition is correct with respect to a particular engagement, but it cannot simply be extended to the design stage. Feasibility is assessed in the circumstances ruling at the time; it is not a licence to create those circumstances in advance. A party that designs a system so that acceptance will never be feasible has not encountered an infeasibility — it has produced one. The question for review is therefore not whether acceptance was feasible on a particular occasion, but whether the system as designed leaves any occasion on which acceptance could be feasible.

Article 57 provides a second route by which the temporal interval enters legal review, without deriving a right to a pause from Article 41. Article 57(2)(b) of Additional Protocol I requires an attack to be cancelled or suspended if it becomes apparent that the objective is not a military objective or is subject to special protection, or that the attack would be disproportionate. A corresponding obligation, qualified by feasibility, is recorded as customary in Rule 19 of the ICRC Study, on the basis of practice extending to States not party to the Protocol. That duty presupposes something about the engagement it governs: where legally relevant information becomes available before the point of irreversibility, the architecture has not already made cancellation or suspension impossible by design.

The proposition is narrow, and its limits should be stated clearly. It does not require every weapon to remain technically recallable after release, nor does it require every engagement to be interruptible on demand. It requires this: where the intended circumstances of use allow a legally relevant change of status to become apparent while the engagement remains causally alterable, the review must examine whether the system preserves an effective pathway by which that information can lead to cancellation, suspension, or referral for further human judgment. The duty to cancel or suspend is not specific to surrender; it applies whenever the legally relevant conditions identified by the rule become apparent. The interval it presupposes is therefore not created for the benefit of a surrendering adversary. Recognition of surrender is one of the cases it serves.

National instruments give this requirement a more concrete operational form. DoD Directive 3000.09 requires autonomous and semi-autonomous weapon systems to be designed so that commanders and operators can exercise appropriate levels of human judgment over the use of force. It also requires engagements to remain within a timeframe, geographic area, and other relevant environmental and operational constraints consistent with commander and operator intentions. If a system cannot remain within those constraints, it is to terminate the engagement or obtain additional operator input before continuing (section 1.2.a.(1)(b), reprised among the matters to be verified before formal development at section 4.1.c.(2)). The same directive requires clear procedures for trained operators to activate and deactivate system functions (section 1.2.a.(3)(c)) and, as an ethical principle of governability, the ability

to disengage or deactivate deployed systems that demonstrate unintended behaviour (section 1.2.f.(5)). These provisions do not create a general entitlement to interrupt any engagement on demand. They establish instead that continuation of an engagement is conditioned in advance on temporal and operational constraints, and that a system unable to remain within them must stop or defer to a human.

Where such a pathway is relied upon, it must be genuinely exercisable. A nominal halting pathway that a co-deployed system can tamper with, falsify, or circumvent does not preserve the temporal opportunity it is supposed to provide. Arrestability must therefore be assessed at the system-of-systems level rather than agent by agent (see HFA Technical Note TN-01, 'Third-Party Interference with Shutdown Mechanisms', 5 July 2026).

The scope of this route should not be overstated. A halting capacity supplies time; it does not supply perception. The fact that a system can be stopped within the cycle does not mean that a manifestation occurring within that cycle will be detected, classified, and acted upon. Those remain separate requirements, and this note does not derive them from arrestability. The narrower point is sufficient for the present argument: the temporal interval whose preservation is at issue is not an indulgence sought on behalf of an adversary, but a property that the deploying party has already accepted for independent reasons of control. Where the interval must be preserved in any event, the question of what can be recognised within it properly belongs before the review.

The consequence for Article 36 is direct but conditional. Article 36 of Additional Protocol I requires a State party, when studying, developing, acquiring, or adopting a new weapon, means, or method of warfare, to determine whether its employment would, in some or all circumstances, be prohibited by applicable international law. Article 36 prescribes that determination; it does not prescribe a particular temporal test. Where, however, the intended circumstances of use allow a person to become *hors de combat* during the engagement cycle, the determination cannot be completed without assessing whether the system preserves a sufficient opportunity for the protected status to be expressed, recognised, and translated into attack inhibition before the engagement becomes irreversible. The review must therefore examine the end-to-end interval the system affords, the conditions under which it does so, and how that interval is measured and recorded. A review that omits this question has not examined whether compliance with the recognition obligation remains possible in the intended circumstances of use. What matters is that the question be reached, not that it be answered in a predetermined way. The outcome may be a design safeguard, an operational restriction, or the exclusion of particular circumstances of use from the authorisation.

The argument does not depend on ratification of Additional Protocol I and can also be stated within the legal architecture of a State not party to it. The United States is the principal example. It is bound by the customary prohibition on attacking persons *hors de combat*, recorded as Rule 47 of the ICRC Customary IHL Study and applicable in both international and non-international armed conflict. Its own military manual recognises the same protection as binding law, extending it to persons in the power of an adverse party, persons incapacitated by wounds or sickness, and persons who have laid down their arms or surrendered (U.S. Department of Defense, Law of War Manual, section 5.9; on surrender, section 5.9.3). The historical root of the rule is Article 23(c) of the Regulations annexed to the 1907 Hague Convention IV. The United States ratified that Convention in 1909, and it entered into force for the United States on 26 January 1910. Addressing the general participation clause in Article 2, the International Military Tribunal at Nuremberg found it unnecessary to decide the issue because, by 1939, the rules in the Convention were recognised by all civilised nations and regarded as declaratory of the laws and customs of war. The obligation relied on here is therefore customary and does not depend on the treaty applying of its own force.

Such a State also conducts legal reviews of new weapons under its own directives. DoD Directive 5000.01 requires legal review of the acquisition of weapons and weapon systems, while DoD Directive 3000.09 requires autonomous weapon systems to be reviewed against national and international law, including the law of war. For that State, the verification identified in this note is therefore not an external requirement derived from a treaty it has declined to ratify. It is an operational consequence of

obligations that the State has itself adopted and has not disavowed. The question is the same under either legal architecture: does the engagement cycle preserve the exchange that the State's own rules presuppose?

This note is the perceptual-relational counterpart to a parallel criterion already deposited: systemic arrestability (Zenodo, DOI 10.5281/zenodo.20837150), which requires that a system's capacity to halt a hazardous action not depend on human perception, evaluation, and intervention within the action's time window. Arrestability concerns the capacity to stop; the reciprocal window concerns the capacity to recognise. Both are temporal criteria attached to obligations already in force, and neither creates new law.

4. Operative conclusion

This note therefore places one question before any State operating or reviewing autonomous or automated engagement systems: if the recognition obligation presupposes a minimum reciprocal time window, and that window is neither guaranteed by design nor examined in the legal review of the weapon, on what basis is compliance asserted? The note does not claim that compliance cannot be established. It records, in a dated instrument, that the question has been asked. The date of deposit records when this specific and testable compliance question was publicly articulated; it does not create the underlying obligation. Nor does the deposit alter the law or place any State on constructive notice. It provides a dated, public, and citable formulation of the proposition that compression of the window is an examinable design parameter rather than merely a technical circumstance.

Subsequent work will develop an empirical demonstrator of the temporal dependence of recognition, showing whether varying the window alone, while holding all other variables constant, changes the outcome of the recognition process.

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